Logan Kojiro

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SUMMARY

Experienced software engineer with a background building verification and testing platforms for complex hardware systems. Seeking engineering roles at the intersection of software and creative products. Foundations in computer graphics, parallel computing, and computer systems.

WORK EXPERIENCE

Ampere Computing — Verification Engineer *Santa Clara, CA | Aug 2021 – Mar 2026*

- Co-built and served as secondary owner of an internal Python coherency-checking tool modeling Ampere's mesh network — adopted by 90%+ of the mesh DV team and run alongside thousands of weekly regression tests, catching hundreds of bugs in the team's proprietary hierarchical coherency model.
- Debugged test failures, working with designers to fix RTL bugs or working with the team to fix bugs in our checkers
- Architected a C++ DPI layer enabling mid-run bug detection in simulation and emulation, letting engineers kill failing runs early, cutting wasted compute on long overnight regressions.
- Contributed to a company-wide quarterly "Kilowatt" award for our team of 3, recognizing the tool's impact on debugging turnaround time.

ManTech — CNO Developer Intern *Herndon, VA | May 2019 - Aug 2019, May 2020 - Aug 2020*

- Built a JavaScript browser-fingerprinting tool that identified and enumerated features of unknown browsers with over 90% accuracy, plus a companion suite of Python tools to verify and extend its functionality.
- Wrote position-independent code in Assembly and C to extract target operating system information from within a browser sandbox.

ARIN Technologies — Software Engineer Intern *Pittsburgh, PA | May 2018 - Aug 2018, Feb 2019 - May 2019*

- Designed and implemented a scalable, user-friendly GUI that cut calibration time for ARIN's indoor positioning system by 95% compared to manual calibration.

CMU BioRobotics Lab — Research Assistant *Pittsburgh, PA | Feb 2017 - Aug 2017*

- Collaborated with an interdisciplinary team of engineers and biologists to emulate and analyze snake locomotion and behavior in a robotic platform.

EDUCATION

Carnegie Mellon University *Pittsburgh, PA | May 2021*

B.S. Electrical & Computer Engineering

PROJECTS

Terminal Tracer — C++20 Real-Time 3D Rasterizer *logankojiro.com*

- Software rasterizer rendering real-time graphics from .obj mesh files as ASCII artwork in the terminal
- Implemented Z-buffering, MSAA anti-aliasing, and Lambertian shading for accurate depth handling and surface lighting.

GPU Fluid Simulation — SPH Solver in CUDA & OpenMP *Parallel Computing and Architecture Final Project*

- Implemented a Smoothed Particle Hydrodynamics fluid solver in three variants — sequential, OpenMP, and CUDA
- Designed a Z-order spatial grid with a GPU-optimized compact variant for optimized neighbor searches
- Documented speedups in a real-time simulation of 100,000+ particles across two machines, observing over 200,000x CUDA speedup compared to our sequential implementation.

TECHNICAL SKILLS

Languages: Python, C, C++, CUDA, Assembly

Real Time & Interactive Systems: Real-time rasterization pipeline, Path Tracing (Monte Carlo Integration), GPU-accelerated rendering, 3D geometry & mesh processing, Inverse Kinematics (Jacobian Inverse)

Systems & Tools: Unix/Linux, Git, OpenMP, Protobuf, CHI & AXI Protocol, Parallel Computing, Cache Coherency, ARM Architecture

LEADERSHIP & ACTIVITIES

Carnegie Mellon ECE Department — Teaching Assistant | Sigma Phi Epsilon Fraternity — Greek Sing Chair, House Manager | Boy Scouts of America — Eagle Scout, Patrol Leader, Instructor